



GENERAL SIR JOHN KOTELAWALA DEFENCE UNIVERSITY

FACULTY OF TECHNOLOGY

INTAKE 42

**BT (HONS) IN INFORMATION AND COMMUNICATION TECHNOLOGY DEGREE
PROGRAMME**

PROFESSIONAL ETHICS (ITIC2153)

TUTORIAL

Activity Overview

Group Size 7–8 Students per Group

Q1. A social media platform deliberately designs its notification system to trigger dopamine responses — sending alerts at calculated intervals to maximize time-on-app, even after internal research showed this worsens anxiety in users under 25. Name TWO specific ethical violations this represents, using precise terminology.

Q2. A fintech startup collects users' precise real-time location every 30 seconds, stores it indefinitely, and sells aggregated movement data to insurance companies — all disclosed in paragraph 47 of a 90-page terms of service document. Identify every ethical principle this violates and explain why each applies.

Q3. A cloud provider runs its primary data centres in a region powered entirely by coal. A green-energy-powered region is available at 12% higher cost. The engineering team chooses the cheaper region to meet quarterly budget targets. What is the ethical failure here and whose responsibility is it?

Q4. During a code review you discover a colleague has copied 400 lines verbatim from a GPL-licensed open source library into your company's closed-source commercial product — with no attribution and no licence disclosure. Your manager says "it's fine, no one will notice." Name TWO specific actions you should take, and explain the ethical and legal consequences of doing nothing.

Q5. Explain the difference between a false positive and a false negative in the context of an AI hiring tool, and explain why minimising one necessarily increases the other. Why is this trade-off an ethical decision rather than a purely technical one?

Q6. A company discovers a zero-day vulnerability in a competitor's widely-used product. A member of the security team suggests holding the vulnerability privately for three months to gain competitive advantage before disclosing it. Explain why this is an ethical violation and state the correct professional course of action.

Q7. An ICT manager instructs their team to ship a product by end of quarter despite knowing three critical security modules are incomplete. The manager tells the team: "We'll patch it in the next release." What specific ethical obligations does this manager violate, and what should a developer on the team do?

Q8. A company trains a facial recognition model on a dataset scraped from public social media profiles without user consent. The model achieves 98% accuracy on light-skinned faces and 74% accuracy on dark-skinned faces. The company argues: "The overall accuracy is excellent." Identify two specific ethical failures in this argument.

Q9. You are an ICT consultant hired to recommend a cloud architecture for a client. One of the vendors has offered you a paid speaking engagement worth \$5,000 at their annual conference. You are about to recommend that vendor's platform. What must you do before making any recommendation, and what happens to your professional standing if you do not?

Q10. A development team builds a mobile banking app that requires users to upgrade to a device released within the last two years — even though the app's functions could run on five-year-old hardware with minor code optimisation. The team made this decision to simplify testing. Name the ethical principle this violates, explain the specific harm it causes, and propose one concrete technical solution.

Answer the following questions

Case Study 1 - The Data Leak That Didn't Happen

Scenario

A software engineer discovers a critical vulnerability in a popular health-tracking app. Before it is exploited, they patch it quietly. Their manager tells them not to disclose the issue publicly to protect company's reputation.

Should the public have a right to know about what could have been leaked?

Is the engineer ethically obligated to report it anyway?

How does responsible disclosure differ from whistleblowing?

Case Study 2 - The Biased Resume Filter

Scenario

A Sri Lankan startup uses a machine learning model to shortlist applicants. Over time, it is found that resumes from certain rural regions are being automatically rejected due to bias in training data.

What ethical obligations does the company have to applicants?

Should the system be discontinued or improved?

How can bias be detected and mitigated early?

Case Study 3 - The App That Tracks Too Much

Scenario

A mental health mobile app collects more data than necessary – including GPS, device microphone usage, and third-party analytics – even though it's only advertised as a journaling tool.

Is this an ethical breach of user consent?

How does this relate to principles of data minimization and transparency?

What design practices can prevent such privacy issues?

Case Study 4 - The Sustainable Software Dilemma

Scenario

An ICT consulting firm proposes building a lightweight, energy-efficient app, but the client demands flashy, GPU-intensive features that consume high bandwidth and energy.

Should engineers push back against client demands in favour of environmental ethics?

What are their professional duties in such a case?

Can aesthetics and sustainability co-exist?

Case Study 5 - The Cloud Migration Whistleblower

Scenario

An employee discovers their company is moving sensitive user data to a third-party cloud server that fails to meet local data protection laws. Management refuses to change providers due to cost savings.

Should the employee report this externally?

What protections might apply to whistleblowers in Sri Lanka?

How can this be handled internally first?

Case Study 6 - The Virtual Reality Playground

Scenario

A VR platform used in education allows avatars to interact freely. Some users report harassment and inappropriate behaviour, but there is no moderation policy in place.

Who is responsible for regulating behaviour in virtual spaces?

What ethical design choices can help prevent misuse?

How does digital identity factor into accountability?

Case Study 7 - The Ghost Developer

Scenario

A developer leaves a freelance project incomplete. Later, their work is sold by the client as a proprietary product with all credits removed.

Does the original developer retain moral rights?

What is the ethical responsibility of the client?

How do intellectual property and fairness apply?

Case Study 8 - The Drone Surveillance Trial

Scenario

A city tests AI-powered drones for traffic and crowd monitoring, but they also collect facial data without public consent. The footage is later shared with private firms for marketing analysis.

Is this an ethical violation of surveillance boundaries?

What role should engineers play in shaping policy?

How does informed public participation matter here?

Case Study 9 - The Internship That Exploits

Scenario

An IT student is asked to build a fully working e-commerce system during an unpaid internship, with promises of exposure and a job that never comes.

When does training cross the line into exploitation?

Should institutions step in to protect students?

How can ethical codes guide internship policies?

Case Study 10 - The AI Artist

Scenario

A tech company launches an AI tool that generates art. It was later discovered that the model was trained on copyrighted work without the artist's consent.

Who owns the rights to AI-generated art?

Is it ethical to profit from work derived from uncredited creators?

Should engineers include consent verification in training datasets?